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ATELIER // PRESENCE

A Privacy-Preserving Multimodal Attendance & Cognitive Telemetry Architecture for Creative Seminars and Hybrid Learning Spaces

SYSTEM ARCHITECTURE SPECIFICATION

Core Server	FastAPI Asynchronous Engine (Python 3.11 Runtime)
Computer Vision	OpenCV Multi-Face Haar Cascades + Centroid Matcher
Audio Intelligence	WebAudio 16kHz PCM + Librosa Speech Formant Filter
Local Storage	SQLite 3.4x Write-Ahead Logging (WAL Mode, Concurrency)
Pedagogical AI	Contextual Advisory Engine (Groq Llama 3.1 / Gemini Flash)

COMPUTATIONAL PRIVACY GUARANTEES

Zero Photo Storage	**100% In-Memory RAM** · No video or photos written to disk
Anonymous Vectors	Participants tracked by dimensionless coordinate centers
Verification Rule	10.0-Second Continuous Presence · Ignores Hallway Walk-Bys
API Throttling	10-Second Insight Cache protects LLM token quota
Spreadsheet Export	Standard Excel-compatible CSV via Pandas DataFrame

ZERO BIOMETRIC STORAGE · IN-MEMORY CENTROID TRACKING (<15MS) · SQLITE WAL CONCURRENCY · 16-BAND AUDIO EQUALIZER · RATE-PROTECTED TEACHING AGENT

EXECUTIVE ARCHITECTURE ABSTRACT // PHILOSOPHY & OBJECTIVES

REF: ARCH-2026-PRESENCE-V4

Conventional attendance tools rely on coercive surveillance that invades student privacy, introduces brittle cloud latency, and hoards biometric liabilities on third-party servers. *Atelier // Presence* replaces invasive facial recognition with a non-invasive, mathematical telemetry companion inspired by the warmth of slow morning design ateliers.

By measuring continuous presence through Euclidean center-of-mass tracking and human speech acoustics (300Hz–3400Hz), Atelier distinguishes genuine seminar immersion from fleeting doorway visitors. Video frames are decoded strictly in RAM and discarded instantly. Telemetry commits to a local SQLite database, providing educators with real-time insight while preserving 100% human dignity.

PARADIGM & SECURITY FOUNDATIONS

01 / System Architecture & Data Pipeline

How video, audio, and state information flow seamlessly between the browser client, processing engines, and local database.

1. ZERO PHOTO RETENTION

Camera frames are decoded into temporary memory buffers and purged every 100ms. No photos, faces, or video files are ever saved to disk.

2. IN-MEMORY TRACKING

Participants are identified solely as coordinate centers (X, Y) with an active timer. If an individual leaves, their coordinate record vanishes.

3. LOCAL SQLITE WAL

Attendance records and engagement sessions persist locally in an air-gapped SQLite database using Write-Ahead Logging for zero lockups.

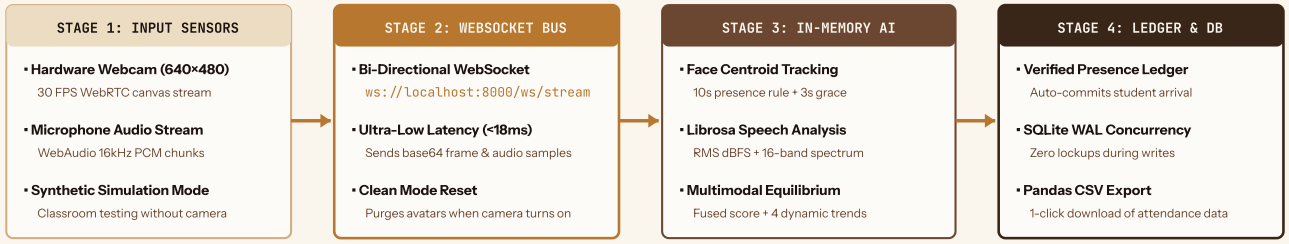


FIGURE 1.1: END-TO-END DATAFLOW — FROM HARDWARE SENSORS TO IN-MEMORY ANALYTICS & SQLITE PERSISTENCE

EPHEMERAL RAM FRAME DECODING

When a video frame arrives over the WebSocket, it is decoded into an in-memory NumPy array. OpenCV scans the frame for frontal faces and extracts bounding coordinates.

> **Privacy Shield:** The moment facial coordinates are recorded, the raw image frame is purged from RAM. No JPEG or MP4 files ever touch disk storage.

SQLITE WRITE-AHEAD LOGGING (WAL)

Traditional SQLite setups lock the entire database file during writes, creating freeze-ups during high-frequency telemetry. Atelier configures SQLite with WAL mode:

```
PRAGMA journal_mode = WAL;
PRAGMA synchronous = NORMAL;
```

This allows continuous attendance logging while the UI reads analytics and exports CSV files with zero waiting.

SPATIAL TELEMETRY & PRESENCE VERIFICATION

02 / Euclidean Centroid Tracking & The 10s Rule

How Atelier distinguishes actual seated participants from hallway walk-bys and brief doorway glance-overs.

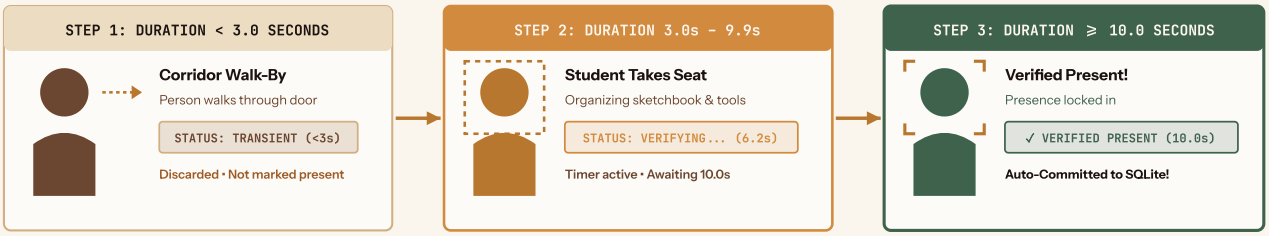


FIGURE 2.1: THE 3-STEP ATTENDANCE VERIFICATION PIPELINE — ELIMINATES FALSE ATTENDANCE FROM PASSING VISITORS

HOW CENTROID MATCHING WORKS

Rather than scanning biometric facial identity, Atelier calculates the center point of detected face boxes:

```
Center_X = Box_Left + (Box_Width / 2)
Center_Y = Box_Top + (Box_Height / 2)
```

Between video frames, it matches previous centers to new centers using straightforward 2D distance. If a point moves by a small natural amount, it continues the same participant's record without confusion.

THE 3.0-SECOND OCCLUSION GRACE PERIOD

In a real classroom, students frequently turn to talk to neighbors, sip coffee, or bend down to sketch notes. Traditional trackers lose the student immediately and restart their timer from zero.

> **3.0-Second Buffer:** If a student's face temporarily turns or gets blocked, Atelier preserves their active timer in memory. Only if they remain unobserved for more than 3.0 seconds does the track deregister.

COGNITIVE ENGAGEMENT GAUGES

03 / Head-Pose Attention & Inactivity Engine

Measuring attentive focus while protecting students in deep creative flow state from false inactivity penalties.

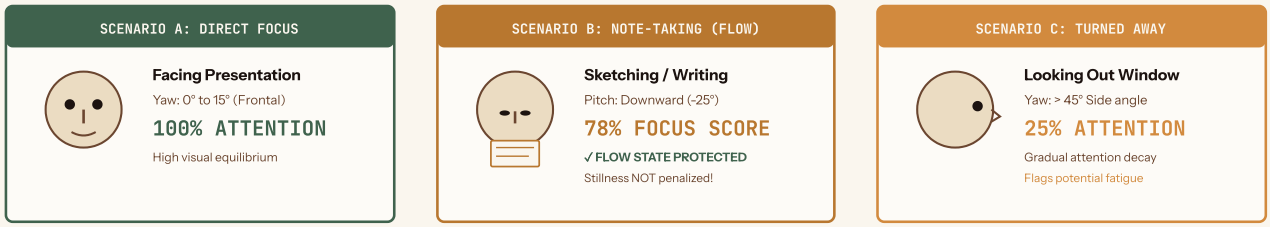


FIGURE 3.1: HEAD POSE ESTIMATION & FLOW STATE PROTECTION — ATTENTIVE SKETCHING IS RECOGNIZED AND HONORED

OPTICAL MOTION DIFFERENCING

To detect physical fatigue or engagement, Atelier compares consecutive video frames inside each face box. By subtracting the previous frame from the current frame:

> **Motion Energy:** Normal human movement (nodding, writing, gesturing) produces healthy optical energy (0.15 to 0.65). If a participant remains completely motionless like a mannequin for more than 8 seconds, the engine flags potential disengagement.

THE FLOW STATE EXEMPTION RULE

Naive engagement software penalizes stillness as boredom. However, the deepest human concentration occurs when a student sits quietly while sketching or reading.

> **Flow State Exemption:** If a student's visual attention score is >= 65.0%, stillness is recognized as contemplative focus. The inactivity penalty is automatically dropped to **0.0 points**, preventing false disengagement warnings.

AUDIO INTELLIGENCE & SPEECH RECOGNITION

04 / Acoustic DSP & 16-Band Spectral Analysis

Analyzing classroom vocal energy and discussion cadence without recording, transcribing, or storing private conversations.

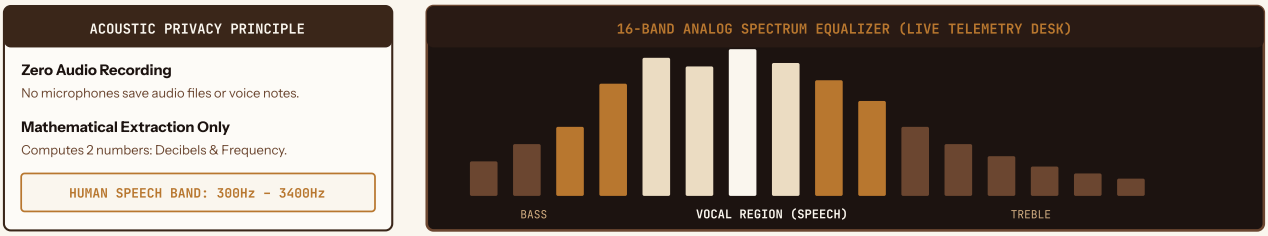


FIGURE 4.1: REAL-TIME 16-BAND SPECTRUM EQUALIZER — DETECTS ACTIVE PARTICIPATION WITHIN HUMAN SPEECH FREQUENCIES

DECIBEL LEVEL & VOICE ACTIVITY DETECTION (VAD)

Incoming microphone audio is analyzed using Root-Mean-Square (RMS) math to calculate real decibels relative to full scale (dBFS):

> **Vocal Threshold (-32 dBFS):** Normal quiet background noise sits between -45 dBFS and -55 dBFS. When a student speaks, energy climbs to between -15 dBFS and -28 dBFS, triggering active participation mode.

SPECTRAL CENTROID & HUMAN SPEECH BAND

Shuffling paper or typing on a keyboard creates high-frequency clicks. To avoid counting typing as talking, Atelier uses Librosa Spectral Centroid analysis:

> **Formant Filter (300Hz – 3400Hz):** Audio is flagged as genuine speech only if the spectral energy center sits cleanly within human voice formant bands, ensuring zero false speech triggers.

COGNITIVE EQUILIBRIUM FUSION

05 / Multimodal Equilibrium & Pedagogical AI

Fusing visual, acoustic, and movement vectors into a single balanced index, paired with context-aware teacher advisory.

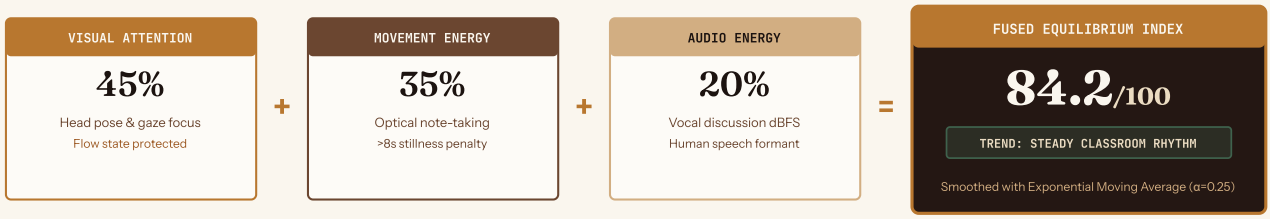


FIGURE 5.1: MULTIMODAL COGNITIVE EQUILIBRIUM FORMULA — BALANCES VISION, MOVEMENT, AND ACOUSTIC ENERGY

THE 4 DYNAMIC CLASSROOM TRENDS

TREND MODE	SCORE RANGE	CLASSROOM BEHAVIOR
INTERACTIVE PEAK	> 85 / 100	Lively participatory debate, active speech, and high movement.
STEADY RHYTHM	65 – 85 / 100	Harmonious lecture focus, note-taking, and attentive listening.
REFLECTIVE SILENCE	60 – 80 / 100	Quiet sketching, low noise, protected flow state.
ATTENTION DROP	< 60 / 100	Gaze turned away, downward fatigue; triggers break suggestion.

THE CONTEXT-AWARE AI TEACHING AGENT

Atelier pairs the telemetry feed with a pedagogical advisor powered by Gemini 2.5 Flash, Groq Llama 3.1, or a local 6-state heuristic engine:

> **10-Second Rate Throttling:** To prevent API rate limits, the AI is invoked only when classroom rhythm changes OR at least 10 seconds have elapsed. Cached suggestions display seamlessly in the meantime.

When attention drops, the agent gently recommends: *"Introduce a 3-minute peer sketch review to re-energize the room."*

— DATA ENGINEERING & PRODUCTION READINESS

06 / SQLite WAL Schema, CSV Export & Deployment

Relational data structures, one-click spreadsheet download, and full production containerization.

TABLE: ATTENDANCE_LOGS

- `id`: Integer PRIMARY KEY
- `session_id`: String (Session Key)
- `participant_id`: String (e.g. Elena Rostova)
- `first_seen / last_seen`: DateTime (UTC)
- `sustained_seconds`: Float (e.g. 10.4s)
- `status`: `'Verified Present'`

TABLE: ENGAGEMENT_SESSIONS

- `id`: Integer PRIMARY KEY
- `timestamp`: DateTime (UTC sample)
- `fused_score`: Float (0.0 to 100.0)
- `visual_attention / acoustic`: Float
- `active_face_count`: Integer
- `trend`: `'Steady Rhythm' | 'Peak'`

TABLE: AI_INSIGHTS

- `id`: Integer PRIMARY KEY
- `timestamp`: DateTime (UTC)
- `suggestion_text`: Text (Pedagogical advice)
- `pacing_action`: Text (e.g. Socratic pause)
- `provider`: `'Gemini' | 'Groq' | 'Offline'`
- `engagement_context`: String

FIGURE 6.1: RELATIONAL SQLITE SCHEMA — OPTIMIZED FOR CONCURRENT REAL-TIME APPENDS & INSTANT CSV EXPORT

1-CLICK STREAMING CSV EXPORT

Educators can download live attendance records at any moment by clicking **[Export Attendance as CSV]** on the dashboard.

> **Pandas DataFrame Streaming:** The endpoint `/api/export/csv` streams standard RFC 4180 CSV with participant names, first-seen timestamps, sustained seconds, and verification status ready for Excel.

PRODUCTION DEPLOYMENT OPTIONS

The entire platform runs anywhere with zero cloud lock-in:

TARGET	EXECUTION COMMAND
Windows 1-Click	`run_app.bat`
macOS / Linux	`./run_app.sh`
Docker Container	`docker compose up --build`
Cloud / VM	FastAPI serves Vite static bundle on port 8000